

EXNO: 01**Setting up Hyperledger Fabric Development Environment****Aim**

To install and configure the Hyperledger Fabric development environment on a Linux system using Docker, Go, and Fabric binaries, create a blockchain test network, and establish a communication channel between peer organizations

By the end of the experiment:

- Hyperledger Fabric installed
- Docker running
- Fabric binaries downloaded
- A blockchain network running
- A communication channel (mychannel) created between two organizations

Procedure:**1. sudo apt install -y git curl jq docker-compose**

this command uses administrative privileges to automatically download and install four developer tools—Git, Curl, JQ, and Docker Compose

2. Verify the installation

```
└─$ git --version
git version 2.53.0

(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ curl --version
curl 8.20.0 (x86_64-pc-linux-gnu) libcurl/8.20.0 OpenSSL/3.6.2 zlib/1.3.2 bro
tli/1.2.0 zstd/1.5.7 libidn2/2.3.8 libpsl/0.22.0 libssh2/1.11.1 nghttp2/1.69.
0 nghttp3/1.22.1 nghttp3/1.15.0 mit-krb5/1.22.1 OpenLDAP/2.6.10
Release-Date: 2026-04-29, security patched: 8.20.0-5
Protocols: dict file ftp ftps gopher gophers http https imap imaps ipfs ipns
ldap ldaps mqtt mqttts pop3 pop3s rtsp scp sftp smtp smtps telnet tftp ws wss
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTP3 HTTPS-proxy IDN I
Pv6 Kerberos Largefile libz PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zst
d

(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ docker-compose --version
Docker Compose version 2.40.3-3

(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ ^C

(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
└─$ docker --version
Docker version 28.5.2+dfsg4, build 9cc6dea35e9a963f281434761c656fba4ac43aed
```

3. Start and enable the docker:

```
(ram@ram)-[~]
$ systemctl list-unit-files | grep docker
docker.service                                enabled
docker.socket                                enabled

(ram@ram)-[~]
$ sudo systemctl enable docker
[sudo] password for ram:

(ram@ram)-[~]
$ sudo systemctl start docker

(ram@ram)-[~]
$ docker info
Client:
Version:      28.5.2+dfsg4
Context:      default
Debug Mode:   false
Plugins:
buildx: Docker Buildx (Docker Inc.)
  Version:  0.29.1+ds1
  Path:     /usr/libexec/docker/cli-plugins/docker-buildx
compose: Docker Compose (Docker Inc.)
  Version:  2.40.3-3
  Path:     /usr/libexec/docker/cli-plugins/docker-compose
```

4. Add yourself to Docker group

```
sudo usermod -aG docker $USER
```

5. Refresh groups

```
newgrp docker
```

6. Test Docker

```
docker run hello-world
```

```
(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
```

7. Check Go

```
sudo apt install golang-go
go version
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.

(ram@ram)-[~]
$ go version
go version go1.26.5 linux/amd64

(ram@ram)-[~]
$ mkdir -p ~/fabric-dev
```

8. Download Hyperledger Fabric

```
curl -sSL https://bit.ly/2ysbOFE | bash -s
```

This downloads:

- Fabric binaries
- Docker images
- Sample applications

```

(ram@ram)-[~/fabric-dev]
$ curl -sSL https://bit.ly/2ysb0FE | bash -s

Clone hyperledger/fabric-samples repo

⇒ Cloning hyperledger/fabric-samples repo
Cloning into 'fabric-samples' ...
remote: Enumerating objects: 15757, done.
remote: Counting objects: 100% (2915/2915), done.
remote: Compressing objects: 100% (554/554), done.
error: RPC failed; curl 56 Recv failure: Network is unreachable
error: 3566 bytes of body are still expected
fetch-pack: unexpected disconnect while reading sideband packet
fatal: early EOF
fatal: fetch-pack: invalid index-pack output
fabric-samples v2.5.16 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.
fatal: not a git repository (or any of the parent directories): .git

Pull Hyperledger Fabric binaries

⇒ Downloading version 2.5.16 platform specific fabric binaries
⇒ Downloading: https://github.com/hyperledger/fabric/releases/download/v2.5.16/hyperledger-fabric-linux-amd64-2.5.16.tar.gz
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
t
                                Dload  Upload  Total  Spent  Left  Speed

```

9. Enter the Sample Folder

cd fabric-samples

10. Configure PATH

```

echo 'export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin' >> ~/.bashrc
source ~/.bashrc

```

11. Verify Installation

Check Peer:
peer version

Check Orderer:
orderer version

```
(ram@ram)-[~/fabric-dev/fabric-samples]  
$ export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin
```

```
(ram@ram)-[~/fabric-dev/fabric-samples]  
$ peer version
```

```
peer:  
Version: v2.5.16  
Commit SHA: f871cf9  
Go version: go1.26.4  
OS/Arch: linux/amd64  
Chaincode:  
  Base Docker Label: org.hyperledger.fabric  
  Docker Namespace: hyperledger
```

```
(ram@ram)-[~/fabric-dev/fabric-samples]  
$ orderer version
```

```
orderer:  
Version: v2.5.16  
Commit SHA: f871cf9  
Go version: go1.26.4  
OS/Arch: linux/amd64
```

12. Start the Blockchain Network

Go into the test network

```
cd ~/fabric-dev/fabric-samples/test-network
```

13. Start it

```
./network.sh up
```



```

(ram@ram)-[~/fabric-dev/fabric-samples]
$ cd test-network

(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
$ ./network.sh up
Using docker and docker-compose
Starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and
  using database 'leveldb' with crypto from 'cryptogen'
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
/home/ram/fabric-dev/fabric-samples/test-network/../../bin/cryptogen
Generating certificates using cryptogen tool
Creating Org1 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org1.ya
ml --output=organizations
org1.example.com
+ res=0
Creating Org2 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org2.ya
ml --output=organizations
org2.example.com
+ res=0
Creating Orderer Org Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-orderer
.yaml --output=organizations
+ res=0
Generating CCP files for Org1 and Org2
[+] Running 7/7
✓ Network fabric_test                                Crea ...           0.0s
✓ Volume compose_peer0.org1.example.com              Created            0.0s
✓ Volume compose_peer0.org2.example.com              Created            0.0s
✓ Volume compose_orderer.example.com                  Created            0.0s
✓ Container peer0.org2.example.com                    Started            0.2s
✓ Container orderer.example.com                       Started            0.2s
✓ Container peer0.org1.example.com                     Started            0.1s
CONTAINER ID   IMAGE                                COMMAND             CREATE
D New Volume   STATUS                                PORTS
NAMES
f7f27b14a5a9   hyperledger/fabric-orderer:latest   "orderer"          Less t
han a second ago   Up Less than a second   0.0.0.0:7050→7050/tcp, [::]:7
050→7050/tcp, 0.0.0.0:7053→7053/tcp, [::]:7053→7053/tcp, 0.0.0.0:9443→944
3/tcp, [::]:9443→9443/tcp   orderer.example.com
a8a53b3de1fa   hyperledger/fabric-peer:latest      "peer node start"   Less t
han a second ago   Up Less than a second   0.0.0.0:9051→9051/tcp, [::]:9
051→9051/tcp, 7051/tcp, 0.0.0.0:9445→9445/tcp, [::]:9445→9445/tcp
peer0.org2.example.com
5d5fabee5bea   hyperledger/fabric-peer:latest      "peer node start"   Less t
han a second ago   Up Less than a second   0.0.0.0:7051→7051/tcp, [::]:7
051→7051/tcp, 0.0.0.0:9444→9444/tcp, [::]:9444→9444/tcp
peer0.org1.example.com
2492821887e5   hello-world                          "/hello"            57 min
utes ago          Exited (0) 57 minutes ago
fervent_dubinsky

```

14. Create Channel

```
./network.sh createChannel -c mychannel
```

This creates a communication channel named mychannel.

It also:

- creates the channel

- joins both organizations

- configures anchor peers

```

(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
$ ./network.sh createChannel -c mychannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb'
Bringing up network
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
[+] Running 3/3
  ✓ Container orderer.example.com      Running      0.0s
  ✓ Container peer0.org1.example.com    Running      0.0s
  ✓ Container peer0.org2.example.com    Running      0.0s
CONTAINER ID   IMAGE                                COMMAND                  CREATE
D              STATUS                                PORTS
NAMES
f7f27b14a5a9   hyperledger/fabric-orderer:latest   "orderer"              21 sec
onds ago      Up 21 seconds                        0.0.0.0:7050→7050/tcp, [::]:7050→705
0/tcp, 0.0.0.0:7053→7053/tcp, [::]:7053→7053/tcp, 0.0.0.0:9443→9443/tcp, [
::]:9443→9443/tcp   orderer.example.com
a8a53b3de1fa   hyperledger/fabric-peer:latest      "peer node start"      21 sec
onds ago      Up 21 seconds                        0.0.0.0:9051→9051/tcp, [::]:9051→905
1/tcp, 7051/tcp, 0.0.0.0:9445→9445/tcp, [::]:9445→9445/tcp
peer0.org2.example.com
5d5fabee5bea   hyperledger/fabric-peer:latest      "peer node start"      21 sec
onds ago      Up 21 seconds                        0.0.0.0:7051→7051/tcp, [::]:7051→705
1/tcp, 0.0.0.0:9444→9444/tcp, [::]:9444→9444/tcp
peer0.org1.example.com
2492821887e5   hello-world                          "/hello"                58 min
utes ago      Exited (0) 58 minutes ago
fervent_dubinsky
Using docker and docker-compose
Generating channel genesis block 'mychannel.block'
Using organization 1
/home/ram/fabric-dev/fabric-samples/test-network/.. /bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/mych
annel.block -channelID mychannel
2026-07-16 05:01:32.908 IST 0001 INFO [common.tools.configtxgen] main → Load
ing configuration
2026-07-16 05:01:32.911 IST 0002 INFO [common.tools.configtxgen.localconfig]
completeInitialization → orderer type: etcdraft
2026-07-16 05:01:32.911 IST 0003 INFO [common.tools.configtxgen.localconfig]
completeInitialization → Orderer.EtcdRaft.Options unset, setting to tick_int
erval:"500ms" election_tick:10 heartbeat_tick:1 max_inflight_blocks:5 snapsho
t_interval_size:16777216
2026-07-16 05:01:32.911 IST 0004 INFO [common.tools.configtxgen.localconfig]
Load → Loaded configuration: /home/ram/fabric-dev/fabric-samples/test-networ
k/configtx/configtx.yaml
2026-07-16 05:01:32.912 IST 0005 INFO [common.tools.configtxgen] doOutputBloc
k → Generating genesis block
2026-07-16 05:01:32.912 IST 0006 INFO [common.tools.configtxgen] doOutputBloc
k → Creating application channel genesis block
2026-07-16 05:01:32.912 IST 0007 INFO [common.tools.configtxgen] doOutputBloc
k → Writing genesis block
+ res=0
Creating channel mychannel
Adding orderers

```



```
(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
```

```
$ ls channel-artifacts
```

```
config_block.json          Org1MSPanchors.tx
config_block.pb            Org1MSPconfig.json
config_update_in_envelope.json Org1MSPmodified_config.json
config_update.json         Org2MSPanchors.tx
config_update.pb           Org2MSPconfig.json
modified_config.pb         Org2MSPmodified_config.json
mychannel.block            original_config.pb
```

```
(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
```

```
$ mychannel.block
```

```
bash: mychannel.block: command not found
```

```
(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
```

```
$ ls organizations/peerOrganizations/org1.example.com
```

```
ca connection-org1.json connection-org1.yaml msp peers tlsca users
```

```
(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
```

```
$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATE
--------------	-------	---------	--------

STATUS	PORTS	NAMES
--------	-------	-------

f7f27b14a5a9	hyperledger/fabric-orderer:latest	"orderer"	4 minutes ago
Up 4 minutes	0.0.0.0:7050→7050/tcp, [::]:7050→7050/tcp, 0.0.0.0:7053→7053/tcp, [::]:7053→7053/tcp, 0.0.0.0:9443→9443/tcp, [::]:9443→9443/tcp	orderer.example.com	

a8a53b3de1fa	hyperledger/fabric-peer:latest	"peer node start"	4 minutes ago
Up 4 minutes	0.0.0.0:9051→9051/tcp, [::]:9051→9051/tcp, 7051/tcp, 0.0.0.0:9445→9445/tcp, [::]:9445→9445/tcp	peer0.org2.example.com	

5d5fabee5bea	hyperledger/fabric-peer:latest	"peer node start"	4 minutes ago
Up 4 minutes	0.0.0.0:7051→7051/tcp, [::]:7051→7051/tcp, 0.0.0.0:9444→9444/tcp, [::]:9444→9444/tcp	peer0.org1.example.com	

```
(ram@ram)-[~/fabric-dev/fabric-samples/test-network]
```

```
$ ls channel-artifacts
```

```
config_block.json          Org1MSPanchors.tx
config_block.pb            Org1MSPconfig.json
config_update_in_envelope.json Org1MSPmodified_config.json
config_update.json         Org2MSPanchors.tx
config_update.pb           Org2MSPconfig.json
modified_config.pb         Org2MSPmodified_config.json
mychannel.block            original_config.pb
```